

LESSON 5 WORKSHEET

Draw the trend, not the dots

Practise best-fit straight lines and best-fit curves

Name _____ Class _____ Date _____

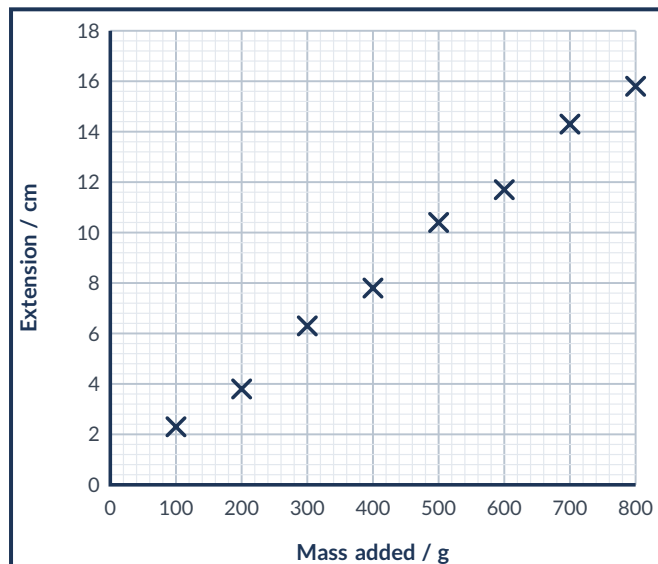
A best-fit line shows the trend, not every measurement.

Straight trend: use a ruler. Curved trend: one smooth freehand curve. Aim for roughly as many points above as below.

1 A straight line

SPRING

A spring stretches as masses are added. Draw the best-fit **straight line** with a ruler.

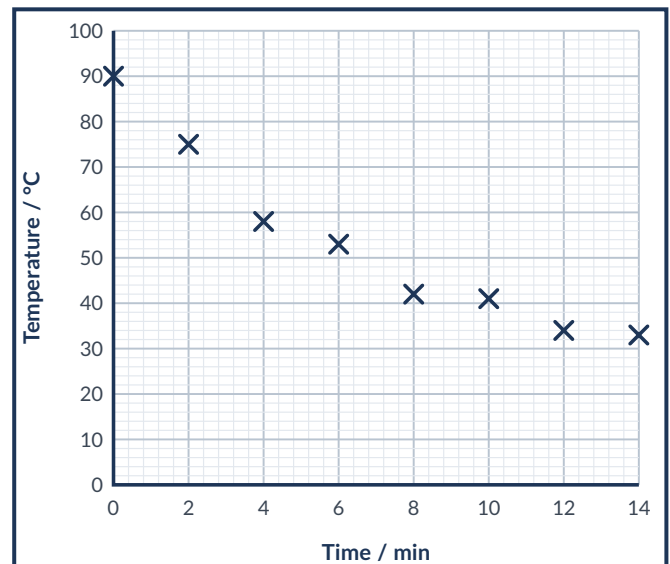


- a Predict the extension for 350 g: _____ cm
 b Points above / below your line: _____ / _____

2 A curve

COOLING

Hot water cools in a room. The trend bends, so draw one smooth **curve**, not a ruler line.

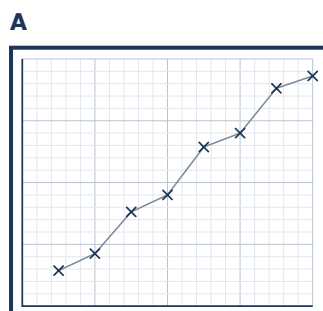


- a Estimate the temperature at 5 min: _____ °C
 b Why is a ruler line a poor choice here?

3 Good line or not?

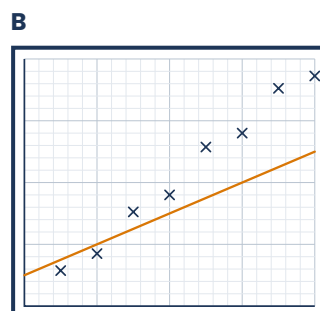
JUDGE IT

Four students drew a line on the same results. Tick **Good** or **Not good** for each and give the reason in a few words.



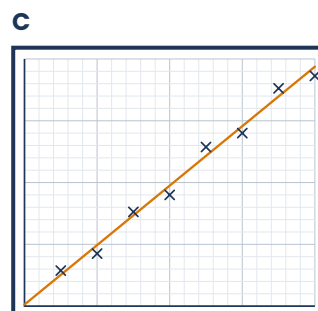
☐ Good ☐ Not good

Why: _____



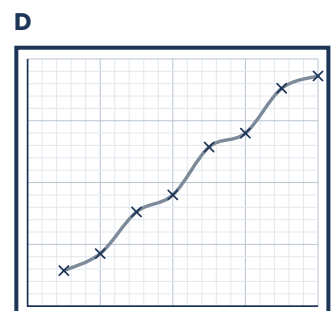
☐ Good ☐ Not good

Why: _____



☐ Good ☐ Not good

Why: _____



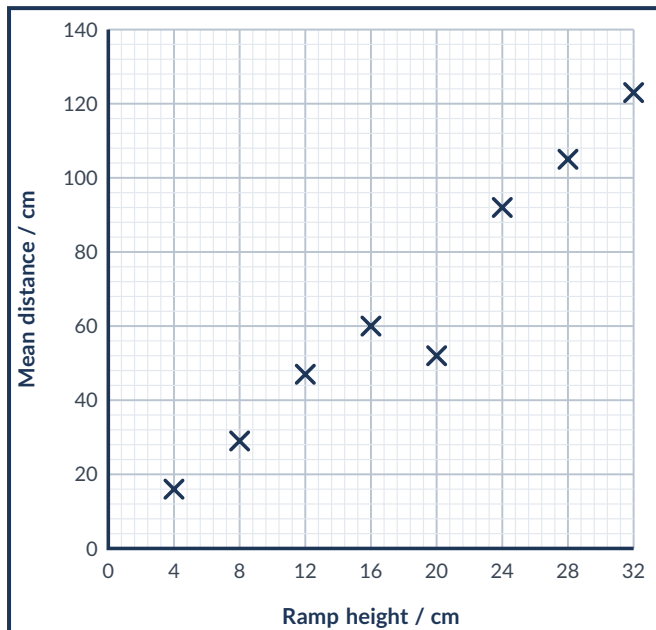
☐ Good ☐ Not good

Why: _____

4 The point that does not fit

OUTLIER

A class repeats the ramp practical and plots the mean distance for each height.



a Circle the possible outlier, then draw the best-fit line for the trend.

b Priya says “rub it out, it spoils the graph”. Marcus says “draw the line through it”. Who is right? What should happen first?

c Repeating 20 cm gives about 76 cm. What does that suggest, and what do you do with the 52 cm reading?

d If it was still 52 cm after repeating, what might that tell you?

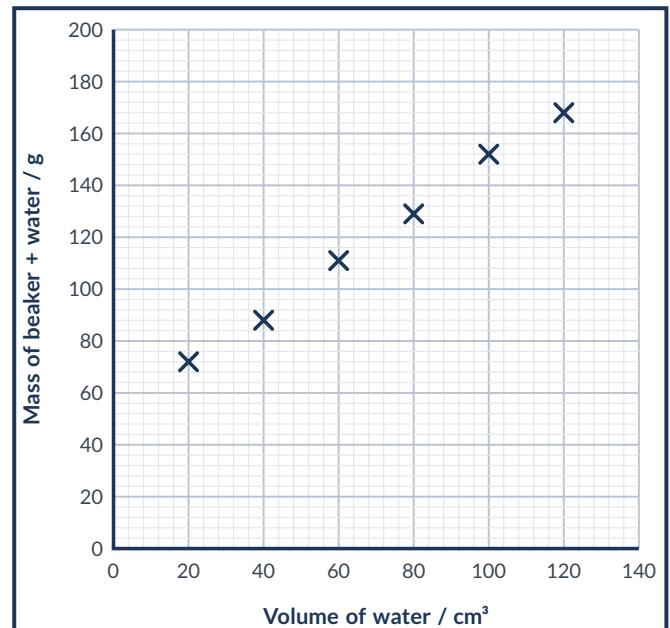
5 Directly proportional?

THINK

DIRECTLY PROPORTIONAL

Double one quantity and the other doubles. On a graph: a **straight line through the origin** (0, 0).

A student pours water into a beaker and measures the **total mass** for each volume. The water alone should be directly proportional to its volume.



a Draw the best-fit line and extend it back to the y-axis. It crosses at _____ g.

b From your line, total mass at 40 cm³ is _____ g and at 80 cm³ is _____ g. Did it double?

c Is total mass directly proportional to volume here? Explain.

d What does the y-intercept represent? How could the student change the method?

TALK ABOUT IT

When, if ever, is it honest to leave an outlier out when you draw a best-fit line?

A straight line with a y-intercept is not directly proportional. Is it still a useful pattern? What can you still predict from it?